

Bhadra Korukonda

Full Stack Developer

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Detail-oriented Computer Science student with hands-on experience building full-stack web applications and production-grade backend systems using Java and Spring Boot; developed scalable REST APIs, React frontends, and end-to-end systems using PostgreSQL, MongoDB, Docker, and CI/CD pipelines. Proven ability to ship clean, maintainable code and optimize performance in real-world projects.

SKILLS

Programming Languages: Python, Java, JavaScript

Libraries/Frameworks: PyTorch, scikit-learn, NumPy, Flask, React

Tools/Platforms: FastAPI, Flask, REST APIs, Git, Docker, Kubernetes, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, SQL

PROFESSIONAL EXPERIENCE

Research Centre Imarat(RCI), DRDO

Intern |

June 2024 – Aug 2024

- Worked on real missile trajectory datasets (lat/long sequences): wrote Python scripts to automatically detect and correct redundancies, fill/interpolate missing values, and clean sequences for direct use in guidance code.
- Designed and delivered a desktop Git client for an intranet environment (no public internet access), enabling scientists to perform add/commit/push/pull operations via the UI instead of using command line arguments. This solution reduced version-control errors and onboarding time for engineers working on classified trajectory and guidance code
- Added SQL-based checks in the CI pipeline to validate cleaned dataset integrity before commits, preventing bad data from ever reaching the repo.

IIT Hyderabad

Research Intern | Python, PyTorch Geometric, Pandas, Geopy, GRU/LSTM, GNNs (GCN, GAT, EGAT) May 2025 – Aug 2025

- Implemented a hybrid Edge-Aware Graph Attention Network (EGAT-GRU) to forecast PM2.5 levels across Delhi using spatio-temporal air quality data.
- Encoded wind direction and geodesic distance as edge features, enabling wind-informed pollutant transport learning and dynamic spatial attention.
- Achieved state-of-the-art performance (RMSE = 16.72, MAE = 10.05), outperforming GAT, GCN, and physics-based baselines for real-time pollution prediction.

PROJECTS

Expense Tracker | Java, Spring Boot 3, Spring Data JPA, Docker, React PostgreSQL, Mongo DB

- Built a complete expense tracking web app to replace messy spreadsheets.
- Used PostgreSQL with Spring Data JPA for all structured data (users, expenses, categories) because transactions and monthly reports need reliable joins and ACID guarantees.
- Used MongoDB for storing receipt images and free-form notes – avoids complex schema changes and makes binary storage simple.
- Implemented standard REST APIs with input validation, pagination, date-range filters, and proper error handling.
- Created a clean React frontend with expense forms, Recharts dashboards (pie/bar charts), search, and responsive layout
- Dockerized backend (Spring Boot) and frontend (React) in separate containers with docker-compose — enables independent builds, faster iteration, and matches real production deployment patterns.

Cloud Automation & Monitoring Pipeline | Docker, Aws, Prometheus, Graphana, Jenkins, Python

- Took a complete 3-tier web application (frontend, backend, database) and fully automated its entire lifecycle.
- Built a Jenkins pipeline that handles Git triggers → Docker builds → Kubernetes blue-green deployment → automated rollback on failure.
- Integrated Prometheus + Grafana dashboards for real-time monitoring of CPU, memory, latency, and deployment status.
- Reduced end-to-end deployment time from several hours of manual steps to a fully automated, one-click process with zero-downtime blue-green deployments.

Automated Resume Ranking System using NLP and MCDM | Python, Flask, spaCy, scikit-learn, PyMCDM, Tesseract OCR

- Built end-to-end resume evaluation system with Flask REST APIs, real-time processing pipeline, and interactive React dashboard.
- Integrated Tesseract OCR + spaCy NLP pipeline to extract and structure resume data, enabling automated candidate scoring and ranking.
- Implemented TOPSIS/AHP algorithms to compute weighted candidate scores, improving shortlisting efficiency by ~40% and reducing human bias.
- Deployed modular, Docker-containerized application allowing HR users to upload resumes, adjust weights, and view ranked results instantly.

EDUCATION

Bachelor of Technology (CS Engg), Woxsen University, GPA: 3.0/4.0

Aug 2022 – July 2026

Narayana Jr College(SSC), Hyderabad, Percentage: 87.1%

June 2020– May 2022

Chinmaya Vidyalaya(CBSE),Hyderabad Percentage: 90%

May 2008 – June 2020